SYSTEM CARD · COMPOUND AI SYSTEM

MAGIK

Multimodal Agentic RAG Integrated Knowledge Assistant

A finance-domain retrieval-augmented generation system that ingests text, PDF, DOCX, XLSX, images, audio and video; routes each query through an agentic controller; retrieves with a hybrid lexical and dense pipeline; verifies its own answers against the retrieved evidence before they reach the user; and runs entirely on self-hosted open-weight models, with no third-party language-model API anywhere in the request path. This edition goes deeper than a typical system card into authentication, tenant isolation, observability and AWS infrastructure, reports the first production measurement of the system, and discloses the real operational and security incidents recorded to date.

MODALITIES	MODELS COMPOSED	CI MERGE GATES	AWS ACCOUNTS TO DATE
7 pipelines, fully isolated	19 open-weight ~43 GB · 0 fine-tuned	1 enforced: retrieval 2 configured, run on demand	3 · each migration forced by GPU quota

Release **v1.0.4** · 6 October 2026 · in production 27 October 2026
Compiled 27 October 2026 · MIT License

Author **Vijaya Karthik**

Contents

01 Overview
02 Architecture
03 Component models
04 Intended use
05 Training and provenance
06 Evaluation
07 Answer verification
08 Guardrails
09 Authentication and session lifecycle
10 Multi-tenant isolation
11 Observability and monitoring
12 AWS infrastructure and deployment
13 Reproducibility
14 Incident disclosures
15 Limitations and known issues
16 Responsible AI considerations
17 References

How to read this card. Every figure is drawn from the repository itself (the model manifest, the evaluation threshold file, the Terraform modules, the monitoring configuration and the test tree), from the release's own Tier-2 report, or from a production measurement made on 27 October 2026. Quality gates that are configured but not enforced, open defects and real incidents (including a security one) are included in Sections 06, 07, 12, 14 and 15 rather than omitted.

01 Overview

MAGIK is built around one premise: most retrieval-augmented generation demonstrations fall apart under the conditions real traffic creates: malformed documents, adversarial input, numbers silently rounded in transit, one user's session bleeding into another's context. This system is an attempt to hold up under those conditions, and to show it with a measured evaluation harness rather than a claim.

It specialises in **financial documents** because finance is an unforgiving domain for retrieval-augmented generation: a hallucinated number is worse than a hallucinated sentence. Financial figures in a generated answer are checked against the literal text of the retrieved chunks, within a 0.5% tolerance. That check is reported on every evaluation run; its 0.95 threshold is configured but not yet enforced as a merge gate (Section 06).

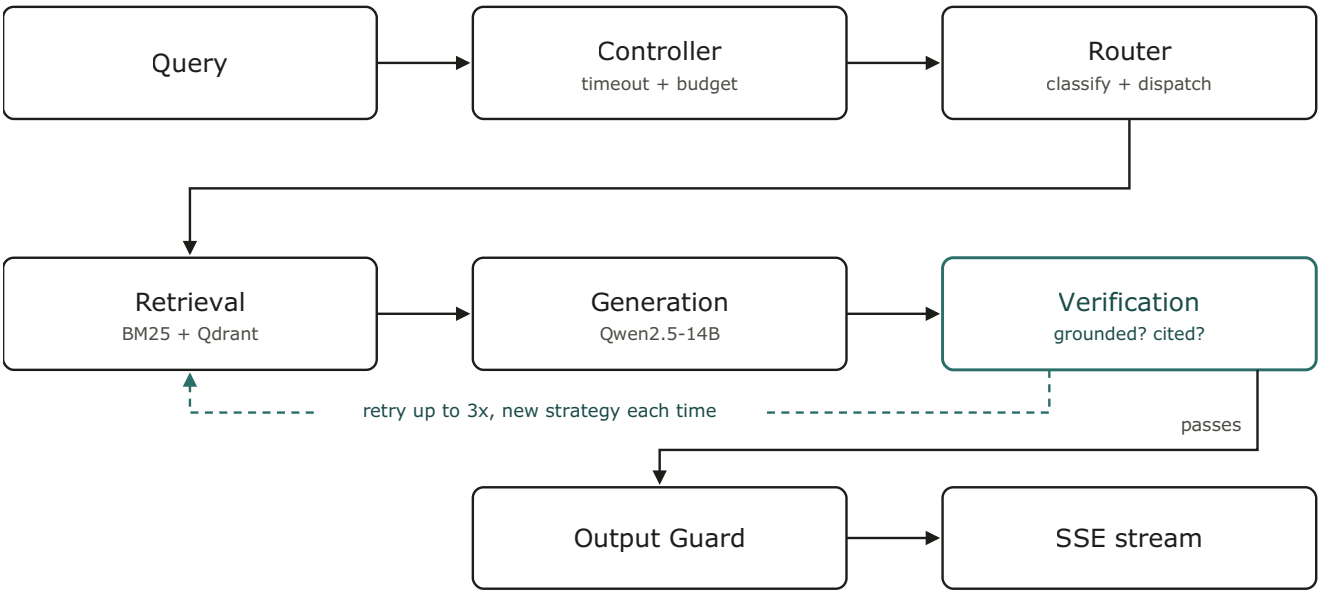
Every model in the stack is open-weight and self-hosted: a quantised GGUF language model served through `llama.cpp`, open embedding and reranking models, and open speech, OCR and vision models. There is no commercial inference API in the request path.

MODALITIES INGESTED	MODELS COMPOSED	TEST FILES	AWS ACCOUNTS TO DATE
7: text, PDF, DOCX, XLSX, image, audio, video	19 · ~43 GB, all open-weight	149 across 8 categories	3, each forced by GPU quota

02 Architecture

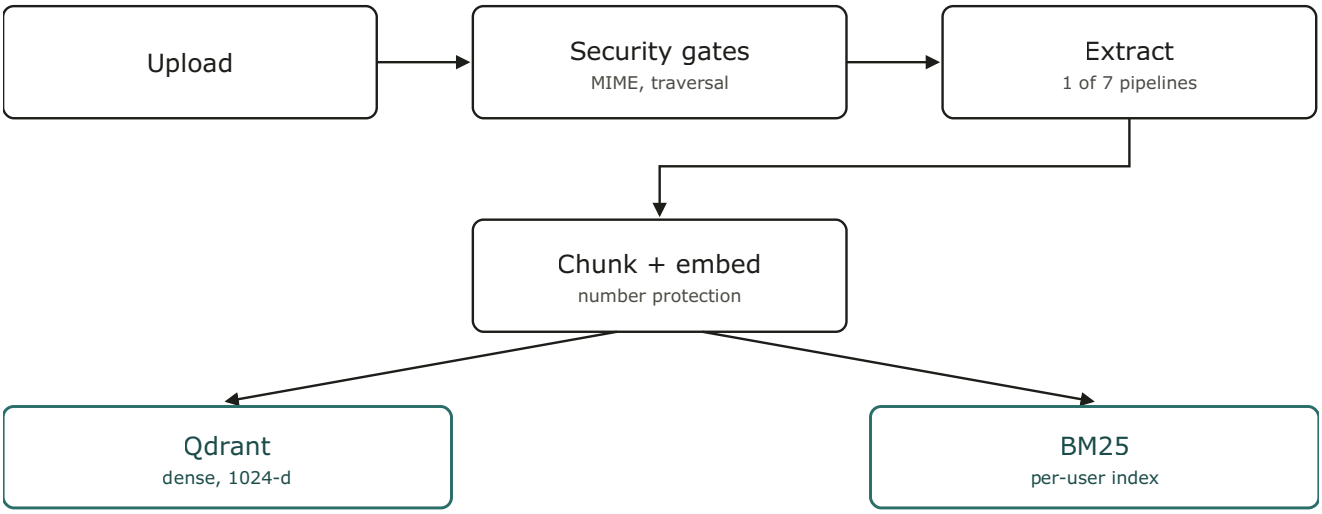
Two independent flows: a query path that answers a question, and an ingestion path that indexes a document. The query path runs under an enforced timeout and token budget; the ingestion path runs security gates before any file content is trusted.

Query flow



The router classifies each query once and dispatches a single tool call: a bounded dispatch, not an open-ended agent loop. A failed groundedness or citation check sends the query back through retrieval with a different strategy each time (up to three retries, bounded by a 120-second timeout). Measured retry behaviour is reported in Section 07.

Ingestion flow



Every ingested file is written to both stores. Uploaded originals and the per-user BM25 index land on the instance's local disk. See Section 14 for why that single-copy design already caused one real data-loss incident.

Per-modality isolation

Every one of the seven modalities owns exactly four files, one per processing layer (ingestion, chunking, embedding, lexical indexing), with no shared per-modality state. This is also what makes the guardrail coverage claim in Section 08 countable: seven modalities across four layers gives 28 text surfaces, each of which must call the single sanitisation entry point.

03 Component models

MAGIK is a compound system rather than a single trained model. Nineteen open-weight checkpoints, roughly 43 GB on disk, are composed through the pipeline above. Sixteen are pinned to an exact upstream commit hash. faster-whisper, Detoxify and EasyOCR are fetched through their own libraries, which have no commit to pin, so they are covered by the checksum alone: the provisioning manifest records a SHA-256 on first download and re-verifies it on every subsequent run. The entailment model used by answer verification was added to the manifest in v1.0.4, so a fresh machine now provisions it.

Provisioned model manifest · all open-weight, none fine-tuned

ROLE	CHECKPOINT	SIZE	PIN
Generation (LLM)	Qwen2.5-14B-Instruct · Q4_K_M GGUF	9.00 GB	commit
Vision, charts and images	Qwen/Qwen2-VL-7B-Instruct	16.59 GB	commit
Vision, video frames	Qwen/Qwen2-VL-2B-Instruct	2.20 GB	commit
Evaluation judge	Qwen2.5-7B-Instruct · Q4_K_M GGUF	4.70 GB	commit
Cross-modal embedding	google/siglip-so400m-patch14-384 · 1152-d	1.76 GB	commit
Speech recognition	Systran/faster-whisper-large-v3	1.55 GB	checksum only
Text embedding	BAAI/bge-large-en-v1.5 · 1024-d	1.35 GB	commit
Reranking	BAAI/bge-reranker-large · cross-encoder	1.34 GB	commit
Answer verification (entailment)	cross-encoder/nli-deberta-v3-base	0.74 GB	commit
Image captioning	Salesforce/blip-image-captioning-large	0.90 GB	commit
Speaker diarization	pyannote/speaker-diarization-3.1	0.60 GB	commit
Financial sentiment	yiyanghkust/finbert-tone	0.44 GB	commit
Named-entity recognition	dslim/bert-base-NER	0.43 GB	commit
Toxicity screening	Detoxify (original)	0.42 GB	checksum only
OCR, printed text	microsoft/trocr-large-printed	0.36 GB	commit
Diarization, segmentation	pyannote/segmentation-3.0	0.20 GB	commit
Diarization, embedding	pyannote/wespeaker-voxceleb-resnet34-LM	0.10 GB	commit
Keyword extraction	all-MiniLM-L6-v2 (KeyBERT)	0.09 GB	commit
OCR, scene text	EasyOCR (en)	0.06 GB	checksum only

04 Intended use

In scope

- Question-answering over finance documents a user has uploaded.
- Demonstrating production-oriented RAG engineering: hybrid retrieval, answer verification, guardrail coverage, tenant isolation, observability and a CI-gated retrieval evaluation.
- Single-GPU, low-concurrency deployment.

Out of scope

- Not financial advice.
- Not evaluated on non-English documents.
- Not load-tested at commercial multi-tenant scale.
- The public demonstration runs a deliberately open shared account.

05 Training and provenance

No model in this system has been fine-tuned. All nineteen checkpoints run at their published open weights: no LoRA, no PEFT, no custom training run anywhere in the stack. The engineering investment goes into retrieval quality, answer verification, guardrail coverage, access control and evaluation rigour around off-the-shelf models.

Text splitting uses `langchain-core` and `langchain-text-splitters` only. The agent's tool routing is a hand-built typed dispatch, not a LangChain agent or chain.

06 Evaluation

Eleven suites are runnable from the harness. **One is enforced as a hard merge gate in CI: retrieval.** A second gate (hallucination) and a third threshold (finance numeric fidelity, minimum 0.95) are written into `thresholds.yaml`, but the hallucination suite is not part of any automated run and the finance threshold is not switched on. Everything else is informational. Earlier editions of this card described all three as enforced; this edition corrects that.

Measurement methodology. Three back-to-back runs on identical code once produced faithfulness scores of 0.363, 0.592 and 0.589. The outlier was the first run after a server restart, not judge stochasticity. Process rule since: discard the first run after any restart, average at least three. Where a confirmation run surfaced a genuinely worse value, a gate is set from the observed maximum, not the average. The release numbers below are single runs, so treat small differences as noise.

Retrieval: the enforced gate CI-gated

retrieval suite · n = 56 · v1.0.4 release gate, run on staging, 27 October 2026 · gate result: exit code 0, no breached sections

METRIC	V5 BASELINE	CI FLOOR	V1.0.4	STATUS
recall@5	0.5089	0.4835	0.6696	pass
MRR	0.3558	0.3380	0.5027	pass
nDCG@10	0.4024	0.3823	0.5564	pass
recall@10	0.5536	0.5259	0.7768	pass
hit rate	0.6786	0.6447	0.9107	pass
context precision	0.0268	0.0255	0.0327	pass

CLOSED SINCE V1.0.2

All six gated metrics clear their floors and improve on the baseline, including the three that were breached in the v1.0.2 edition (recall@5, MRR, nDCG@10). The cause of that recovery has not been attributed, so it is reported as a measurement, not a fix. Retrieval latency in this run: p50 0.023 s, p95 0.029 s, p99 0.223 s.

Hallucination: configured gate, not run automatically

hallucination suite · n = 97, all 7 modalities · last gated run 2026-08-17 (v8) · compared with the v1.0.4 generation suite, n = 98

METRIC	V8 (AUG)	GATE	V1.0.4 RUN	MEANING
fabrication rate	0.0619	max 0.079	0.1531	Primary safety signal
hallucination rate	0.2234	max 0.246	0.3367	Blended metric
omission rate	0.1822	none	0.2653	Completeness signal, no gate

The fabrication gate is set at 10% above the observed maximum across four runs (0.0722). The fourth confirmation run surfaced a real, intermittent hallucination that three identical runs had missed, root-caused and not a metric artifact. The gated hallucination suite has not been re-run for v1.0.4. The v1.0.4 column comes from the informational generation suite, which computes the same metrics over 98 queries. If the gate applied to those values it would fail (0.153 against 0.079, and 0.337 against 0.246). The gap is open and unattributed.

Finance numeric fidelity: threshold configured, not enforced

Financial figures cited in an answer are matched against the literal text of retrieved chunks within a 0.5% tolerance and no unit-scale bridging. The configured minimum is 0.95. Measured: 0.8342 over 98 queries in the v1.0.4 Tier-2 run, and 0.90 over 30 questions in the production run below. Neither meets the configured minimum, which is why the threshold is described as not enforced and not as passing.

Generation and routing: informational

v1.0.4 Tier-2 run on staging, 27 October 2026

METRIC	V1.0.2 CARD	V1.0.4	N
Answer correctness	0.7083	0.6403	98
Answer relevancy	0.6528	0.6352	98
Context recall (deterministic)	0.8962	0.7544	87
Faithfulness (judge)	0.5146	0.4253	97
Finance fidelity (0.5% tolerance)	0.8266	0.8342	98
Citation accuracy (heuristic)	n/a	1.0000	83
Template leak rate	n/a	0.0000	98
Route accuracy, routing suite	1.000	0.7500	20

Not like-for-like: the v1.0.2 values covered 42 text, PDF and DOCX queries; the v1.0.4 run uses 98 queries across the suite's full question set. Route accuracy is 0.75 on the 20-question routing set (15 correct; the five misroutes were direct to rag once, hybrid to search twice, memory to rag once and rag to search once), below the configured floor of 0.917, which is not enforced. The 144-question end-to-end suite reports route accuracy 0.8611.

Other suites in the same run

- **Behavioural suite** (local 7B judge, informational): adversarial pass 0.1429 (n = 14), refusal accuracy 0.4643 (n = 21). Both are weak and are being examined; neither blocks a release.
- **End-to-end suite:** 20 requests were rejected with HTTP 400, and its retrieval-style metrics (hit rate 0.17) disagree with the dedicated retrieval suite (0.91). They are not used in this card.
- **OCR, audio and video suites** produced no scores: the evaluation corpus files were missing on the staging instance (14 questions each).

Production measurement, 27 October 2026

The first measurement taken on the live production instance (AWS g6e.xlarge, one NVIDIA L40S with 46 GB VRAM; Qwen2.5-14B-Instruct Q4_K_M served by llama.cpp fully on the GPU; Qwen2.5-7B-Instruct Q4_K_M as a local judge, no hosted model APIs; 7 finance documents, 1,236 indexed chunks for the evaluation tenant). It ran in its own memory-capped container, so the live app was not disturbed.

Real-time latency · streaming endpoint · 48 questions, 46 answered (2 refused, 0 errors)

MEASURE	P50	P95	MEAN	SAMPLES
Time to first token	11.55 s	23.70 s	13.16 s	46
Full answer time	12.61 s	24.69 s	14.16 s	46

Method: 48 real gold-set questions sent one at a time to `/rag/query/stream`, the endpoint the web UI uses, after two untimed warm-up requests. Answer cache off, each question scoped to its source document and run in its own chat session. Time to first token includes retrieval,

generation and verification, because the server verifies an answer before it streams. Only answered requests are timed. Single user, warm models, no concurrent load.

MODALITY	QUESTIONS	TTFT P50	TTFT P95
text	8	9.05 s	19.41 s
pdf	8	12.50 s	31.81 s
xlsx	6	15.19 s	18.34 s
docx	8	11.52 s	14.26 s
image	8	8.40 s	12.18 s
audio	4	11.79 s	15.11 s
video	4	18.75 s	23.28 s

Per-modality p95 values rest on very few samples and are indicative only.

Answer quality · RAGAS · first 30 answerable gold-set questions (14 text, 14 Apple 10-K PDF, 2 Word); spreadsheet, image, audio and video not included

METRIC	RESULT	COVERAGE	HOW TO READ IT
Context recall	0.93	30 questions	Share of reference facts found in retrieved context
Citation accuracy	1.00	16 answers with citations	Citations point to retrieved sources (heuristic check)
Template leak rate	0.00	30 questions	No prompt text leaked into any answer
Finance number fidelity	0.90	30 questions	Figures in answers match source text verbatim (0.5% tolerance)
Answer relevancy	0.73	see limits below	Question is addressed by the answer
Faithfulness	0.84	11 of 30 gradeable	Statements supported by context; 19 rows the judge could not grade are excluded
Hallucination flag rate	0.33	30 questions	Heuristic flag, 10 of 30 answers flagged; under review

Answer quality · DeepEval · first 10 gold-set questions, all on the text file · raw values, three of six metrics are lower-is-better

METRIC	RAW SCORE	BETTER IS	ROWS SCORED	READING
Answer relevancy	0.40	higher	10 of 10	Low; answers only partly address the question as judged
Faithfulness	0.59	higher	7 of 10	3 rows failed on judge output format
Contextual recall	1.00	higher	9 of 10	1 row failed on judge output format
Hallucination	0.87	lower	10 of 10	High value as returned; not consistent with the RAGAS result
Bias	1.00	lower	9 of 10	Not credible: judge output did not match the expected format
Toxicity	1.00	lower	10 of 10	Not credible: judge output did not match the expected format

WHAT THESE NUMBERS DO NOT SHOW

- **Judge limits.** Scores come from a local 7-billion-parameter model, not human review. It sometimes returns output the scoring library cannot parse, which drops that question from a metric. The report does not yet record how many questions each RAGAS metric scored, so answer relevancy and context recall may rest on fewer than 30.
- **Faithfulness** covers only 11 of 30 answers, so it is a partial figure.
- **Hallucination flag rate** is a rule-based check, not a verified error rate. A five-question smoke run flagged none and this run flagged one third; the flagged answers have not been reviewed.
- **DeepEval reliability.** With a local 7-billion-parameter judge, DeepEval's bias, toxicity and hallucination scores are not credible: a toxicity of 1.00 would mean every answer was toxic, and the judge's replies did not match the format the library expects. Treat those three as unresolved and the other three as indicative. RAGAS and DeepEval figures use different question subsets (30 against 10) and different judge prompts, so they are not directly comparable.
- **Long sessions.** An earlier latency run put all 48 questions in one chat session and 17 were refused; with a separate session per question only 2 were refused (both on the spreadsheet). Many questions about different documents in one long session is a known weak spot under investigation.
- **Sample size.** 30 quality questions and 48 latency questions are a first production measurement, not a benchmark.

Reproduce: `deploy/aws/scripts/run_quality_report.sh ragas 30` and `... deepeval 10` (release v1.0.4); latency from the streaming benchmark described above. Raw reports are kept on the instance under `/opt/magik/quality-reports/`.

Per-modality scorecard: single-run snapshot (2026-08-20)

Taken before v1.0.2 and not repeated per modality for v1.0.4, so read these as history, not as current values. Image is the strongest modality across the board; audio is the weakest.

generation quality · n = 14 / modality

MODALITY	FAITHFULNESS	ANSWER RELEVANCY	ANSWER CORRECTNESS	CONTEXT RECALL
Text	0.4545	0.7045	0.6591	1.0000
PDF	0.5714	0.7679	0.8571	0.2846
DOCX	0.6429	0.8077	0.7857	0.7667
XLSX	0.6154	0.7143	0.7321	0.7917
Image	0.8036	0.8393	0.8929	1.0000
Audio	0.3929	0.6429	0.6786	0.8000
Video	0.3929	0.8214	0.7857	0.8738

hallucination and safety · n = 14 / modality

MODALITY	HALLUC. RATE	FABRIC. RATE	OMISSION RATE	TEMPLATE LEAK	CITATION ACC.
Text	0.0909	0.0909	0.0909	0.0000	1.0000
PDF	0.5000	0.1429	0.3571	0.0000	N/A*
DOCX	0.4286	0.0714	0.3571	0.0000	1.0000
XLSX	0.1429	0.0000	0.1429	0.0000	1.0000
Image	0.0714	0.0714	0.0000	0.0000	1.0000
Audio	0.1429	0.0714	0.1429	0.0000	1.0000
Video	0.2857	0.0000	0.2857	0.0000	1.0000

* PDF's heuristic citation-accuracy metric had zero measurable rows in that run, so it is excluded rather than shown as a false 0 or 100.

verification loop · n = 14 / modality

MODALITY	GROUNDING	CITATION V2	RETRY SUCCESS	AVG RETRIES
Text	1.0000	0.8889	0.5000	0.2222
PDF	0.9286	0.7857	0.3333	0.4286
DOCX	1.0000	0.6923	0.0000	0.1538
XLSX	0.9167	1.0000	0.0000	0.0833
Image	0.9286	0.9286	0.0000	0.0714
Audio	0.7692	0.5385	0.2727	0.9231
Video	1.0000	0.9286	0.0000	0.3571

finance fidelity and latency · n = 14 / modality

MODALITY	FIN. FIDELITY	VER. P50	VER. P95	GEN. P50	GEN. P95	GEN. P99
Text	0.9091	4.01s	11.62s	11.81s	34.07s	44.15s
PDF	0.8433	7.13s	13.95s	9.70s	19.36s	20.52s
DOCX	0.8125	3.91s	9.02s	6.69s	10.91s	11.53s
XLSX	0.9286	4.37s	9.18s	10.20s	14.86s	16.65s
Image	0.9929	3.14s	4.89s	4.33s	6.05s	6.54s
Audio	0.6310	9.21s	15.34s	12.89s	19.29s	22.57s
Video	0.9643	3.97s	6.92s	8.57s	11.03s	11.19s

- Faithfulness was the weakest quality axis system-wide in that snapshot (0.39 to 0.80).
- Audio needed real improvement: weakest citation grounding (0.5385) and lowest finance fidelity (0.6310).
- PDF and DOCX showed elevated hallucination rates (50% and 43%) despite strong correctness.
- Text had unexplained tail-latency spikes (p95 34.07 s, p99 44.15 s) despite being structurally simplest.

07 Answer verification

Before an answer reaches the user, a verification loop checks whether its claims are supported by retrieved context and whether its citations point at real chunks. On failure it retries with a different strategy each time, up to three retries: expand retrieval to twice the top-k, rewrite the query, deepen to three times the top-k, then decompose the query into aspects. The loop never repeats a strategy and is bounded by a 120-second timeout. Groundedness combines a lexical numeric guard with an entailment model (`cross-encoder/nli-deberta-v3-base`).

v1.0.4 Tier-2 run on staging, 27 October 2026 · n = 92 answers

GROUNDING SUCCESS	CITATION ACCURACY (V2)	VERIFICATION LATENCY	MEAN RETRIES / QUERY
0.9239 floor 0.891	0.7935 floor 0.816	p50 5.17s p95 14.35s	0.3696 retry success 0.094

REGRESSIONS TO NOTE, V8 TO V1.0.4

Against the v8 baseline: grounding success 0.9384 to 0.9239 (still above its 0.891 floor); citation accuracy 0.8587 to 0.7935 and retry success 0.186 to 0.094, both below their configured floors (0.816 and 0.177); verification latency p50 2.44 s to 5.17 s. These floors are not enforced gates, and the runs are not like-for-like (v8 averaged three runs; this is a single run on the staging instance), so the movement is reported, not explained.

From the v8 baseline: across 140 retried sessions, 88.6% changed nothing, 8.6% raised the score and 1.4% flipped a failing answer to a passing one. The loop is a safety net that rarely fires usefully, not a general quality multiplier. In the v1.0.4 run, 3 of 32 retried answers eventually passed.

A bug this instrumentation caught. Plain-text documents never triggered the verification loop at all, because ingestion tagged chunks "text" while chunking used the canonical "txt". Fixed by aliasing the two tags.

08 Guardrails

All ingested content is treated as untrusted data, never as instructions. A single entry point is called on every one of the 28 modality-by-layer text surfaces before that text reaches a model.

measured 27 October 2026 on the v1.0.4 input guard · shipped 109-prompt red-team corpus

ATTACK PROMPTS BLOCKED	EXPECTED-PASS PROMPTS BLOCKED	INJECTION PATTERNS	OWASP LLM TOP 10
84 / 84 (100%)	1 / 25 (stale label, see below)	46, severity-tiered	10 / 10 addressed (per MODEL_CARD.md)

The corpus holds 84 prompts that must be blocked and 25 that must pass; each was run through the query-surface input guard, offline on CPU. All 84 attacks were blocked. One expected pass (`enc-002`) was also blocked: it is "ignore previous instructions" hidden inside right-to-left override characters, labelled safe before the guard became Unicode-tolerant. Blocking it is correct, so the label is stale, not the guard.

Earlier editions quoted "64 / 64 recall, 0.9% false positives, F1 0.994". Those figures cannot be reproduced from the corpus as shipped (84 attack prompts, not 64), so the direct measurement replaces them. Only the query surface was measured here; ingestion surfaces and the output guard have their own tests. 118 guardrail test functions cover the same surfaces, and the alerts in Section 11 watch them live.

09 Authentication and session lifecycle

Authentication is a full implementation, built request-flow by request-flow. Every default below is the code's actual default in `app/core/config.py`.

Password login

`POST /auth/login` verifies the password against an Argon2 hash (time_cost=3, memory_cost=65536 KiB, so 64 MiB per verification), with bcrypt as a fallback verifier for legacy hashes.

TOKEN	DEFAULT LIFETIME	CONFIG KEY
Access token	30 minutes	ACCESS_TOKEN_EXPIRE_MINUTES
Refresh token	7 days	REFRESH_TOKEN_EXPIRE_DAYS

Multi-factor authentication (TOTP)

Enrollment generates a standard RFC 6238 TOTP secret plus 8 single-use backup codes, each a 10-character hex string stored only as a bcrypt hash. Verification allows a plus-or-minus 30 second clock-skew window. Login with MFA enabled is two-step: a correct password issues a short-lived MFA token, not the real access token; the client then submits a TOTP or backup code, which only then issues the real JWT pair.

Google OAuth 2.0

Authorization-code flow with CSRF protection via a server-held state parameter, not PKCE. The state store is a bounded in-memory dictionary (capped at 500 entries).

Password reset (OTP)

A 6-digit OTP is stored in Redis with a 10-minute TTL and emailed. 3 wrong guesses trigger a 15-minute lockout. A successful verification issues a one-time reset token (1-hour TTL) consumed on the actual password change.

Logout and revocation

JWTs are stateless, which normally makes logout theatre. On logout or password change, the token's `jti` is written to Redis with a TTL set to the token's own remaining lifetime: the blacklist entry expires at the same moment the token would have anyway, and never grows unbounded.

Rate limiting

Per-user, fixed 60-second window (default 60 requests), independent of the OTP-lockout mechanism above.

10 Multi-tenant isolation

Every data layer filters on the user identifier independently. This is defence in depth: a bug in the API's authorization check does not, by itself, leak data, because the storage layer underneath enforces its own filter regardless of what called it.

DATA LAYER	ISOLATION MECHANISM	FAILURE MODE IF MISSING
Qdrant	Typed field condition on user_id per query	Cross-tenant semantic search
BM25	Per-user index file path	Cross-tenant keyword search
Redis	Namespaced keys under user:{id}:*	Session and conversation bleed
MongoDB	Every query filters on user_id	Reading another's transcript

The public demo is a single shared tenant, not a bypass of isolation. What is not isolated is durability: the demo tenant's uploaded files live only on local disk, which is why demo access can delete the corpus (Section 14). That is a data-durability gap, not a tenant-isolation one.

11 Observability and monitoring

A seven-service stack, defined as code in `docker-compose.monitoring.yml`.

SERVICE	ROLE
Prometheus + Pushgateway	Per-modality counters on a dedicated port; Pushgateway for short-lived batch jobs
OTel Collector	Receives distributed trace spans
Tempo	Stores traces; backs the TraceQL alert below
Grafana	Dashboards and unified alerting, behind Caddy
Loki + Promtail	Log aggregation, correlated by trace ID
Arize Phoenix	LLM observability, bound to 127.0.0.1 only
Uptime Kuma	Passive push from wake and idle-stop Lambdas; never polls (would wake the GPU)

Dashboards

Three, provisioned as JSON: `system_health.json`, `rag_quality.json`, `logs.json`. Since v1.0.3 the RAG Quality dashboard carries an "Offline judge: RAGAS / DeepEval" row: scores per metric, the graded fraction of each run, whether the Qwen judge produced them, last-run age and status, and the evaluation container's peak memory. A failed run never overwrites the last good scores.

Quality report on production (v1.0.3)

`run_quality_report.sh` runs RAGAS and DeepEval on the production box in their own container, under a hard memory cap (12 GB by default) and with the highest OOM score, so if it outgrows its budget the kernel stops that container and nothing else. It refuses to start unless the GPU has room for the judge and the host has room for the cap. It holds a self-expiring `magik:eval-busy-until` instance tag so the idle-stop Lambda does not stop the box mid-run. The evaluation borrows the serving app's retriever and encoder through internal routes that are off by default, return 404 to anything that came through Caddy, and serve only the evaluation tenant.

Alerting: 12 rules across 3 groups

GROUP	ALERT
magik-system	Circuit breaker OPEN
magik-system	Ingestion error rate spike
magik-system	p95 latency breach (online)
magik-system	GPU VRAM critically low
magik-system	Reranker p95 latency breach
magik-system	App error log rate spike (Loki)
magik-system	Span error rate spike (Tempo/TraceQL)
magik-security	Guardrail block rate spike
magik-security	Auth failure spike (login + MFA)
magik-rag-quality	Online hallucination rate drift
magik-rag-quality	Production traffic drift: WARNING
magik-rag-quality	Production traffic drift: CRITICAL

Disclosed rather than hidden. The TraceQL span-error-rate rule requires Grafana v12.1+ and its own feature toggle. Grafana itself documents TraceQL alerting as experimental. Included anyway, with that tradeoff accepted explicitly.

12 AWS infrastructure and deployment

Three accounts, one recurring cause

ACCOUNT	ERA	WHY IT ENDED
537557168406	Original	Superseded during v1.0.0 bring-up
857194222592	v1.0.0 to v1.0.1	GPU vCPU quota of 4: no room for prod + staging
266901698137	v1.0.2 to current	Live account

Every migration shares the same root cause: a g6e.xlarge needs 4 GPU vCPUs, and a fresh account's default quota is exactly 4, enough for one box, never two. The previous module is kept only as an as-built record, marked `DEPRECATED.md`, and is no longer applyable.

The AZ capacity fight (2026-09-08 to 2026-09-12)

Production missed us-east-1a, then 1b, and landed in 1c. Staging then missed every availability zone in the region for three straight days before also landing in 1c. Only the Uptime Kuma box stayed on the original AZ.

Network and access

INSTANCE	INBOUND	RATIONALE
Production	SSH (admin CIDR), HTTP (ACME), HTTPS (app + /grafana/)	Public app, break-glass SSH only
Staging	Zero inbound rules	SSM only: no port to scan, no IP to target

IAM is split by trust boundary: a GitHub OIDC provider lets Actions assume a deploy role with no long-lived credentials in the repo; a separate EC2 role is what the running instances use. CI identity and runtime identity are never the same principal.

Staging to gate to promote pipeline

INSTANCES	STAGING ACCESS	PROMOTION GATE	STAGING UPTIME
2 x g6e.xlarge, identical	SSM only	Tier-2 suite; blocks on retrieval	Minutes per deploy

The production image is built exactly once, deployed to staging, and the Tier-2 suite runs against it. Only if that gate passes does the identical image get promoted to production. The gate can fail only on a retrieval regression: every other sub-suite reports informationally (Section 06). A failing gate rolls staging back and production is skipped entirely, since nothing was ever deployed there. The image build attaches an SBOM and provenance attestation.

CURRENT STATUS: STAGING EXISTS

Staging exists again in the production account; the GPU quota wait recorded in v1.0.2 has ended. It is started only for a tagged deploy and stopped when the gate finishes. The v1.0.4 release ran all of this in one CD run on 26 and 27 October 2026: deploy to staging, the Tier-2 gate (exit code 0, no breached sections), promotion to production, and stop-staging. The self-hosted staging runner had to be re-registered first (Section 14).

Production wake and idle-stop

ALWAYS-ON COST	SCALE-TO-ZERO COST	COLD START	TRANSPORT
~$1,340 / month	~$12 / month + hourly	60 to 90 s wake, then ~7 min model load	HTTPS via Caddy

Since v1.0.4 a cold start shows a "MAGIK AI is starting up" page (Caddy answers 502, 503 and 504 with a page that refreshes every 15 seconds) so the roughly seven minutes of model loading no longer looks like a failed deploy. The start-up script also confirms from llama-server's own log that the 14B model is on the GPU; if the GPU has dropped out of the container it retries, then exits non-zero so the container restarts and the fault is visible. CPU serving remains available only as a deliberate setting.

A bug this separation caught. Staging was once found emitting "env": "production" on every log line: a deploy step layered an environment file that never defined ENV on top of an AMI clone that already had it set.

13 Reproducibility

- **Model weights:** 16 of 19 pinned to an exact upstream commit hash; all 19 SHA-256 verified on every run.
- **Python dependencies:** fully locked.
- **Infrastructure:** codified in Terraform, including the account migration in Section 12.
- **Vector data:** Qdrant collections can be snapshotted and restored.
- **Container images:** built once per release with an SBOM and provenance attached.
- **Offline loading:** pinned models leave a cache entry without a `refs/main`, which the offline app could not resolve. v1.0.4 writes the missing reference at start-up and after every download, repairing an existing cache without a re-download.

This mattered three times: strict manifest checking revealed the 7B vision model had been silently absent from provisioning; `app/bin/restore_demo_kb.py` exists precisely because uploaded files are not yet part of this pinned set (Section 14); and the offline reference fix removed a failure that hit the entity recogniser, image captioner and diarizer on every boot.

14 Incident disclosures

Dated, root-caused, and kept here rather than only in CHANGELOG.md.

2026-09-08 · Knowledge-base wipe during AWS account migration

What happened. Uploaded documents and BM25 indexes live only on an instance's root EBS volume, blank on a newly built box. Migrating production replaced the instance and took the demo corpus with it. Managed stores (Qdrant, MongoDB Atlas, Upstash Redis) were unaffected.

Fix. `restore_demo_kb.py` re-uploads through the normal upload route rather than copying files directly, rebuilding the disk copy, vectors and BM25 index together.

Residual risk. A repair tool, not an architecture fix: the same loss recurs on the next instance replacement (Section 15).

2026-09-08 · Chat transcripts lost a turn on every refused answer

What happened. A refused answer was never written to the transcript, but the frontend patched "the last assistant message" regardless, overwriting the previous turn's real answer while the current question never appeared. An empty knowledge base (the state above) makes nearly every answer a refusal, so one incident silently escalated into a second: history corruption.

Fix. The patch is now turn-aware: a mismatched patch is refused rather than blindly applied, and the missing turn is stored instead of discarded. Covered by a dedicated integrity test.

2026-09-12 · PRIVATE SSH KEY COMMITTED TO A PUBLIC REPOSITORY

What happened. Four Terraform plan files were tracked in this public repo. A .tfplan is a zip whose tfstate members are not redacted the way console output is, so each embedded the full PEM of the RSA-4096 break-glass SSH key for the then-live AWS account.

Why two controls both missed it. The .gitignore pattern matched tfplan3.out but not destroy.tfplan: one wrong glob. CI's detect-secrets gate skips binary content, and a .tfplan is a zip: a blind spot in the tool's design.

Impact, assessed rather than assumed. One RSA key leaked, for an account that no longer exists. No SSM parameter values were exposed, only IAM policy ARNs. The current account generates its own independent key pair; fingerprints were compared directly and confirmed different.

Fix, and a deliberate non-fix. Both glob spellings are now excluded everywhere. The files are untracked but not scrubbed from git history: force-rewriting a public repo's history was judged to trade a real disruption for no real security gain, since the key opens nothing that still exists.

2026-10-02 · Live demo failed: the language model was silently running on CPU

What happened. A GPU that dropped out of the container during start-up left every layer of the 14B model on four CPU cores, and each answer ran into the 630-second generation deadline while the log still said `device=cuda`. On the same day: models pinned to commit hashes failed to load offline (the entity recogniser, image captioner and diarizer failed this way on every boot); the conversational rewrite could return Chinese text; and the roughly seven-minute cold start looked like a failed deploy.

Fix (v1.0.4). The model is served from the GPU or not at all: the start-up script confirms GPU use from llama-server's own log, retries, then exits non-zero. The missing cache references are written at start-up. A rewrite containing CJK text the original lacked is rejected, and the verification limitation notice is appended verbatim, never reworded. A start-up status page replaces the blank error. The NLI groundedness model is now provisioned by the downloader.

Residual risk. The cold start still takes about seven minutes; it is now explained to the visitor, not shortened.

2026-10-26 to 27 · Release-day findings for v1.0.4

What happened. The first v1.0.4 tag landed on a commit that had no CI runs, so the deploy's wait-for-CI step timed out and the tag was moved. GitHub had removed the self-hosted staging runner after it was offline for more than 14 days, leaving the Tier-2 gate job queued until the runner was re-registered. The hosted vector database cluster had been suspended and was resumed with its data intact. A git "dubious ownership" error made the monitoring-sync step fail silently, and a missing IAM permission for the evaluation busy-tag had to be applied through Terraform.

Impact. No data was lost and no incorrect release was promoted; the gate that passed was the one for the final tag. Sources: the CD run history and operator notes.

15 Limitations and known issues

Documented deliberately rather than left implicit.

- **Only retrieval is an enforced merge gate.** The hallucination gate is configured but its suite is not part of any automated run, and the finance numeric-fidelity threshold (0.95) is not switched on (Section 06).
- **Generation-suite fabrication and hallucination rates (0.153 and 0.337) exceed the ceilings the gated suite uses** (0.079 and 0.246). Unattributed; the gated suite has not been re-run for v1.0.4.
- **Verification metrics regressed against the v8 baseline** in the v1.0.4 run: citation accuracy (v2) 0.7935 and retry success 0.094 sit below their configured floors. Informational.
- **Routing accuracy is 0.75** on the 20-question routing set, below its configured 0.917 floor, which is not enforced.
- **OCR, audio and video suites produced no scores** in the v1.0.4 gate run (corpus files missing on the staging instance); the end-to-end suite had 20 rejected requests.
- **Judge reliability.** The local 7B judge returns unparseable output for some rows: RAGAS faithfulness covers 11 of 30 answers, the report does not record scored-row counts per metric, and DeepEval bias, toxicity and hallucination scores are not credible.
- **Production latency is not fast.** Median time to first token is 11.55 s and median full answer 12.61 s (single user, warm models), because an answer is verified before it streams.
- **Cold start takes about seven minutes** after a wake (Section 12).
- **Uploaded files and BM25 indexes have no second copy:** repaired by a restore script, not closed (Section 14).
- **Retrieval context precision is low in absolute terms** (0.033 in the v1.0.4 run).
- **A real, intermittent hallucination remains open** on dense audio transcripts.
- **The hybrid web route does not execute a live web search** (recorded at v1.0.2; the v1.0.3 and v1.0.4 changelogs report no change).
- **Streaming and production evaluation is manual,** not continuous.
- A modality-tagging audit is incomplete beyond the text-tag fix.
- No formal fairness or bias audit has been conducted.
- Single production instance, with no horizontal scaling or failover.
- The TraceQL alert rule runs on functionality Grafana calls experimental (Section 11).

16 Responsible AI considerations

- **No external inference provider:** document and query content never leaves the deployment boundary.
- **PII detection** via Microsoft Presidio across ingestion and memory surfaces.
- **Toxicity screening** on model output via Detoxify.

- **Untrusted-content discipline:** documents, transcripts and web results are always data, never instructions.
- **Auditability:** guardrail violations and auth/MFA failures are monitored and logged.
- **Answer traceability:** citations trace a cited number back to source text.
- **Incident transparency:** real incidents disclosed with root cause and impact (Section 14).

17 References

ITEM	DETAIL
Repository	github.com/vjkarthik98/MULTIMODAL-AGENTIC-RAG-INTEGRATED-KNOWLEDGE-AI-ASSISTANT
Release	v1.0.4 · 6 October 2026 · in production 27 October 2026
License	MIT
Author	Vijaya Karthik
Engineering log	Full change history in CHANGELOG.md, including root-caused production incidents
Measured evaluation	MAGIK_Measured_Evaluation_v1.0.4.pdf (production measurement, 27 October 2026)
Deprecated AWS module	As-built historical record only: deploy/aws/terraform/DEPRECATED.md

Every figure in this card was read from the repository at release v1.0.4, from that release's Tier-2 report, or from the 27 October 2026 production measurement. The attack-prompt measurement in Section 08 was run directly on the v1.0.4 code. Figures that could not be verified were omitted rather than estimated. Where documentation disagreed with code, the code was treated as authoritative.